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Forensic Science 35

KEY

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Summary

	Total Possible Marks	Your Marks
Lesson 1	23	
Lesson 2	26	
Lesson 3	23	
Lesson 4	28	
TOTAL	100	

Teacher's Comments

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Forensic Science 35
Assignment Booklet 6
Anthropology and Forensic Entomology



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Assignment Booklet 6

Forensic Science 35

Assignment Booklet 6

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Module 6, Lessons 1 - 4

Total
23

Lesson 1 Assignment: Determining Ancestry and Sex from Skeletal Remains

Value
10

Part A: Multiple-Choice Questions

Decide which of the choices best completes the statement or answers the question. Print the letter that corresponds to your choice on the blank next to the question.

- A 1. Forensic anthropology is an applied specialized subfield of anthropology in which the goal is to analyze and interpret
- A. human remains
 - B. ancestral groups
 - C. animal skeletons
 - D. ethnic backgrounds

Use the photograph below to answer the next question.



- D 2. The bones shown above are likely
- A. leg bones
 - B. arm bones
 - C. human ribs
 - D. animal ribs

- A** 3. When compared with an animal leg bone, a human femur has a relatively
- A. spherical head
 - B. large trochanter
 - C. short neck between the head and the shaft
 - D. wide region between the head and the trochanter
- B** 4. Which part of the human skeleton tends to exhibit the most distinct traits among individual ancestral groups?
- A. Ribs
 - B. Skull
 - C. Pelvis
 - D. Femur
- C** 5. The two most useful regions of the skull in determining ancestry are:
- A. the maxillary suture and the nasal region
 - B. the orbits and interorbital area
 - C. the orbits and nasal regions
 - D. the interorbital area and the nasal region
- A** 6. The base of the nasal area in a Mongoloid skull is often
- A. flared
 - B. round
 - C. sloped
 - D. rectangular
- C** 7. When compared to other ancestral groups, the space between the eyes and nose in the skull of a Negroid individual tends to be
- A. flared
 - B. sloped
 - C. wider
 - D. smaller

- D** 8. Which of the following three skeletal parts have characteristics that differ in appearance in human males compared to human females?
- A. Ear, spine, teeth
 - B. Ear, femur, spine
 - C. Pelvis, ribs, teeth
 - D. Femur, pelvis, skull

Use the following information to answer the next question.

A hiker found an unknown human skeleton in a ravine. A forensic anthropologist noted the following characteristics of the skull:

- Round chin
- Sloped Orbits
- Large nasal spine
- Small canine teeth
- Slight supraorbital ridges in forehead

- C** 9. The skeleton described is likely the remains of a
- A. Negroid male
 - B. Mongoloid female
 - C. Caucasoid female
 - D. Caucasoid male
- A** 10. When compared to the human male pelvis, the human female pelvis tend to
- A. be wider
 - B. be more narrow
 - C. have more notches and ridges
 - D. have more muscle attachment sites

Value
4

Part B: Numeric-Response Questions

Use the following photographs to answer the next question.

Various Types of Bones**Bone #1****Bone #2****Bone #3****Bone #4**

②

1. Match each bone numbered above with its correct name shown below

Bone photo:	<u>3</u>	<u>1</u>	<u>4</u>	<u>2</u>
Name of bones:	rib	pelvis	femur	skull

Use the following information to answer the next question.

Cranial Features of the Human Skull

1. Round chin
2. Square chin
3. Slight supraorbital ridges
4. Extreme supraorbital ridges
5. Sloped orbits of the eye
6. Rectangular orbits of the eye
7. Large nasal spine
8. Wide interorbital region

②

2. The cranial features numbered above that would most likely be found in the skull of a Negroid male include 2, 4, 6, and 8.

(Record your answers in ascending order.)

Value
9

Part C: Written-Response Questions

Provide appropriate answers to the following. Explanations are best given in sentences.

④

1. Identify four pieces of information that a forensic anthropologist would include in the osteobiography of a subject.

Any four of the following are acceptable:

- **Human vs. non-human remains**
- **Sex**
- **Ancestry**
- **Number of individuals (that is, a mass grave)**
- **Time since death**
- **Stature or height**
- **Age**
- **Evidence for cause of death**
- **Type of injuries, pathologies, genetic abnormalities**

- ③ 2. Describe three ways that human skeletal remains can be differentiated from animal skeletal remains.

Any three of the following are acceptable:

- *Humans normally have 12 pairs of ribs; animals have more than 12.*
- *Human rib bones tend to be C-shaped. Animal bones tend to be linear and/or straight.*
- *The large upper arm bone (humerus) of a human has a smaller tubercle on the upper region.*
- *The large upper arm bone (humerus) of a human has fewer, smaller projections extending from its upper part than animal bones have.*
- *Human arm bones tend to be smaller and more slender than animal limb bones.*
- *The large upper leg bone (femur) of a human has a large, spherical head and a long neck compared to an animal leg bone, which has a wider, multiple-notched upper head portion.*
- *animal femurs have a substantially larger trochanter due to larger muscle attachment sites because joint is positioned more sideways.*

- ② 3. Describe two ways that the sex of skeletal remains can be determined when no skull has been found.

Any two of the following are acceptable:

- *The female has a wider and shorter pelvic region than the male.*
- *The upper leg bone or femur of a human male has a number of large ridges where muscles attach.*
- *Male femurs have larger weight-bearing surfaces*

Total
26**Lesson 2 Assignment: Determining Traumatic Injuries from Skeletal Remains****Value**
10**Part A: Multiple-Choice Questions**

Decide which of the choices best completes the statement or answers the question. Print the letter that corresponds to your choice on the blank next to the question.

- A** 1. When trying to determine the type of traumatic injury that has occurred, the most useful skeletal evidence is
- A. fractures
 - B. ridges
 - C. bone colouration
 - D. bone density
- D** 2. The type of bone fracture that occurs is dependent upon the
- A. race and height of the victim
 - B. age and weight of the victim
 - C. time at which force is applied
 - D. direction from which force is applied
- B** 3. What is the most common type of force that causes traumatic injury?
- A. Tension
 - B. Bending
 - C. Shearing
 - D. Compression
- D** 4. A force that pushes inward from the end of a bone and causes numerous fractures radiating from the point of impact is known as
- A. torsion
 - B. bending
 - C. shearing
 - D. compression

- A** 5. Often when a person holds out his or her arms in self-defence, he or she will suffer a
- A. parry fracture
 - B. bone displacement
 - C. shearing fracture
 - D. bone dislocation
- B** 6. An injury caused by a blow from a large instrument with a flat or round surface is known as
- A. projectile trauma
 - B. blunt force trauma
 - C. sharp force trauma
 - D. strangulation trauma
- D** 7. Which of the following observations describes the bone injury that is most often caused by projectile trauma?
- A. Fractures over a large surface area
 - B. Fractures over many different parts of the skeleton
 - C. Puncture wounds in bone with grazing cut marks from the point of impact
 - D. Displacement of bone with radiating fracture lines from the point of impact
- C** 8. Which of the following objects is most likely to cause projectile trauma?
- A. Brick
 - B. Knife
 - C. Bullet
 - D. Hammer
- B** 9. What will appear in bone if the force during sharp force trauma is perpendicular?
- A. Parry fracture
 - B. Puncture wound
 - C. Bone displacement
 - D. Comminuted fracture

- C 10. Which of the following bones is often fractured in victims of strangulation?
- A. Jaw
 - B. Skull
 - C. Hyoid
 - D. Femur

Value
4

Part B: Numeric-Response Questions

Use the following photographs to answer the next question.

Diagrams of Directions of Force that cause Bone Fractures

Diagram #1



Diagram #2

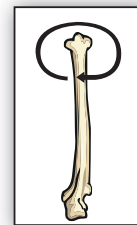


Diagram #3

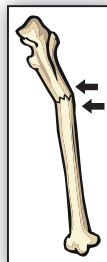
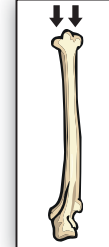


Diagram #4



②

1. Match each bone numbered above with the appropriate type of force given below.

Force Diagram:	<u> 3 </u>	<u> 4 </u>	<u> 1 </u>	<u> 2 </u>
Type of force:	bending	compression	tension	torsion

Descriptions of Various Types of Injuries Found in Skeletal Remains

1. A 2 cm puncture wound in the pelvis
2. Multiple fractures of the hyoid bone
3. Multiple fracture lines over a wide surface area of the skull
4. A 3 cm hole with radiating fracture lines in the breastbone

②

2. Match each bone numbered above with the appropriate type of force given below.

Injury description: 4 2 3 1
Type of trauma: **projectile** **strangulation** **blunt force** **sharp force**

Value
12

Part C: Written-Response Questions

Provide appropriate answers to the following. Explanations are best given in sentences.

③

1. Why is the cause of death usually inferred by forensic anthropologists rather than confirmed after they have studied the skeletal remains?

Many factors may be unknown in deaths caused by some type of trauma. If a stab wound to the torso is found, existence of this type of wound does not actually prove that it was the actual cause of death. Hypothetically, the person could possibly have died from other causes before or after he or she was stabbed.

9

2. Complete the chart below by stating one type of weapon that could have caused each type of trauma, one type of force that produces each type of trauma, and the type or types of bone injury resulting from each type of trauma.

Each box = 0.5 mark. Each cell in columns 2 and 3 requires one answer for 0.5 mark.

Type of Trauma	Type of Weapon	Type of Force	Type(s) of Bone Injury
<i>Blunt Force</i>	<i>club, hammer, boot, bat, rock, brick</i>	<i>compression, bending, shearing</i>	<i>fractures over wide area</i>
<i>Projectile</i>	<i>bullet, arrow, spear</i>	<i>compression, bending</i>	<i>displacement of bone with radiating fractures from impact point</i>
<i>Sharp Force</i>	<i>axe, knife</i>	<i>compression, shearing</i>	<i>puncture wounds or grazing marks</i>
<i>Strangulation</i>	<i>hands, rope</i>	<i>compression</i>	<i>fractured hyoid bone</i>

Total
23

Lesson 3 Assignment: The Use of Forensic Entomology in Criminal Investigations

Value
10

Part A: Multiple-Choice Questions

Decide which of the choices best completes the statement or answers the question. Print the letter that corresponds to your choice on the blank next to the question.

- D 1. Forensic entomology is a useful technique that can help determine
- A. how many victims have been buried
 - B. the role that insects play in the environment
 - C. the number of insects that have colonized a gravesite
 - D. how long human remains have been lying undetected
- D 2. Insects help to recycle organic matter in the environment through the process of
- A. infestation
 - B. estimation
 - C. maturation
 - D. decomposition
- C 3. The colonization of human remains at a predictable rate and within a predictable sequential order is known as
- A. infestation
 - B. decomposition
 - C. insect succession
 - D. insect competition
- D 4. Why does insect colonization of a corpse occur in a predictable way?
- A. Various insects are able to exist during each season.
 - B. Maggots tend not to move as they feed upon a body.
 - C. Female flies tend to lay their eggs in the facial region.
 - D. Different insects feed upon different parts of the body.

- C** 5. An estimation of the age of the insects that have colonized the body can lead to
- A. an inference of the cause of death
 - B. a prediction of the sex of the victim
 - C. an estimation of the time since death
 - D. a determination of the number of victims
- B** 6. Which of the following factors has the greatest effect on the rate of insect activity at a crime scene?
- A. Number of eggs laid
 - B. Environmental temperature
 - C. Distance between the crime scene and a water source
 - D. Distance between the crime scene and the location of the body
- A** 7. The answer that correctly identifies the first type of insect to infest human remains and the second type of insect to infest a dead body is row
- A. Blowflies, Beetles
 - B. Eggs, Larvae
 - C. Beetles, Blowflies
 - D. Pupae, Maggots
- C** 8. In which of the following body regions is a blowfly most likely to lay her eggs?
- A. Arm
 - B. Foot
 - C. Mouth
 - D. Abdomen
- B** 9. When a victim's body is found during the winter, which of the following observation is evidence that the individual died **before** the winter began?
- A. Only one type of insect is found on the body.
 - B. Only dead insect eggs are found in or on the body.
 - C. A small number of maggots are found feeding on the body.
 - D. A large number of blowflies and beetles are found feeding on the body.

- A** 10. When a heavy infestation of eggs and/or maggots are found in an area of the body where there is no natural body opening (such as arms or legs), it may be assumed that
- A. an open wound is located at this site
 - B. the victim's body was lying in the shade
 - C. the death occurred in the spring or summer
 - D. a sexual assault occurred before the victim was killed

Value
4

Part B: Numeric-Response Questions

Use the following information to answer the next question.

Observations of Insects upon Human Remains in Various Types of Conditions

1. A great variety of insects
2. Only beetle infestation
3. Fly eggs and no maggots
4. Insect succession has taken a relatively long time

①

1. Match each of the observations above with the correct type of conditions as given below.

Observation:	<u> 1 </u>	<u> 3 </u>	<u> 4 </u>	<u> 2 </u>
Type of conditions:	sunny	autumn	shade	completely buried

Use the following information to answer the next question.

Various Stages of Maturity of the Blowfly

1. Egg
2. Adult
3. Pupae
4. Larvae

①

2. The correct sequence, as shown above, in which blowflies mature is **1** , **4** , **3** , and **2** .

Use the following information to answer the next question.

Various Observations made by a Forensic Entomologist

1. Tend to feed upon dead tissue that has relatively little blood in it
2. Region of a human corpse that insects tend to first infest and colonize
3. Poisons and drugs can be found in these after they have fed on human remains.
4. Region of a human corpse that insects first infest if a sexual assault has occurred

②

3. Match each of the observations above with the correct term given below.

Observation:	<u>2</u>	<u>3</u>	<u>4</u>	<u>1</u>
Term:	head	maggots	anogenital	beetles

Value
9

Part C: Written-Response Questions

Provide appropriate answers to the following. Explanations are best given in sentences.

①

1. How can insects be used to determine if an individual has suffered from poisoning?

Any maggots or pupae that have fed upon the body can be tested for poisoning.

Use the following information to answer the next question.

The skeletal remains of an adult male were found under some tree branches and snow on the outskirts of Edmonton in January. A forensic entomologist observed more blowfly pupal membranes below the ribs in the chest region than in the skull of the victim.

②

2. a. Use your knowledge of insect maturation or development to explain that the victim described above was deposited in the above location before or after winter began.

It has been observed that during the autumn and winter in the Edmonton area, blowfly eggs do not mature into larvae or pupae. Because pupal membranes were found in the chest region of the victim, significant blowfly development had occurred. Therefore, this had to have happened while the body was lying there in the summer or spring. (more likely the summer or decomposition would have progressed even further).

②

- b. What does the observation of numerous pupal membranes in the chest region suggest about the cause of death of the victim described above?

Numerous pupal membranes in the chest region suggest that the victim likely suffered from some trauma to the chest (such as knife or bullet wound) because blowflies lay their eggs in open wounds more so than in body openings.

②

3. a. What type of database did the Canadian Police Research Centre begin in 1996?

The Canadian Police Research Centre in 1996 began a database about insect succession patterns observed in each of the biogeoclimatic zones in Canada.

②

- b. Why is such a database necessary in Canada?

The database is necessary because insect species and weather conditions vary in the various biogeoclimatic zones across Canada.

Total
28**Lesson 4 Assignment: Crime Case Studies Involving Forensic Anthropology and Forensic Entomology****Value**
12**Part A: Multiple-Choice Questions**

Decide which of the choices best completes the statement or answers the question. Print the letter that corresponds to your choice on the blank next to the question.

Case Study #1: Bones that were Worth a Thousand Words

- B** 1. The faces of the victims were damaged by rifle butts. The type of trauma that this would have caused to the skull of each victim is

A. projectile trauma
B. blunt force trauma
C. sharp force trauma
D. strangulation trauma

- C** 2. Which of the following rows identifies the types of trauma caused by both the bullet and the bayonet wounds that each victim suffered?

Row	Trauma Cased by Bullet	Trauma Caused by Bayonet
A.	Sharp force	Blunt force
B.	Comminuted	Projectile
C.	Projectile	Sharp Force
D.	Blunt force	Strangulation

- D** 3. What did DNA testing of the skeletal remains confirm?
- A. Two skeletons belonging to the Romanov family were missing.
B. Three skeletons were male and six skeletons were female.
C. Nine skeletons suffered from multiple bullet and bayonet wounds.
D. Five skeletons were part of the same family and four skeletons were unrelated.

- A** 4. Three parts of each skeleton that would have helped forensic anthropologists to determine the sex of the victim are the
- A. femur, the pelvis, and the skull
 - B. hyoid, the ribs, and the upper arm
 - C. head region, the trochanter, and the tubercle
 - D. interorbitals, the orbitals, and the nasal region

Case Study #2: The Girl, the Army Sergeant, and the Bugs

- B** 5. What types of trauma would have been evident on the skull of the victim?
- A. Sharp force trauma and projectile trauma
 - B. Blunt force trauma and sharp force trauma
 - C. Projectile trauma and strangulation trauma
 - D. Strangulation trauma and blunt force trauma
- A** 6. Blowflies most likely would have laid their eggs in or near the victim's
- A. eyes and ears
 - B. head and neck
 - C. hair and abdomen
 - D. chest and anogenital region
- A** 7. The insect evidence indicated that the victim had likely died on
- A. May 31st
 - B. June 1st
 - C. June 3rd
 - D. June 4th
- B** 8. What type of insect evidence indicated that this victim had been dead for approximately 4 to 5 days?
- A. Beetles
 - B. Maggots
 - C. Blowfly eggs
 - D. Blowfly pupae

Case Study #3: The Farmhouse Murder

- C** 9. Which of the following groups of skeletal features indicate that the victim was a female?
- A. Forehead, ribs, and skull
 - B. Nasal area, orbitals, and pelvis
 - C. Teeth, pelvis, and supraorbital ridges
 - D. Hyoid, lower arms, and anogenital region
- D** 10. The parry fractures observed in the lower arm bones indicate that the victim was
- A. injured as a child
 - B. sexually abused as a child
 - C. dismembered using a sharp instrument
 - D. trying to defend herself from her attacker
- B** 11. The fractures in the hyoid bone of the victim likely indicate that the victim was
- A. stabbed
 - B. strangled
 - C. dismembered
 - D. sexually assaulted
- D** 12. The blowfly pupae membranes found in the anogenital area of the victim indicate that the victim was
- A. stabbed
 - B. strangled
 - C. dismembered
 - D. sexually assaulted

Value
5

Part B: Numeric-Response Questions

Case Study #1: Bones that were Worth a Thousand Words

Use the following list to answer the next question.

Example Pieces of Evidence from Romanov Remains

1. Height
2. Shape
3. Face shape
4. Dental plate
5. Unique brow shape
6. Mitochondrial DNA
7. Deformed pelvis due to horseback riding
8. Wide, broad, and flat pelvis due to traumatic injuries

②

1. The four pieces of evidence numbered above that helped to identify the skeletal remains of Czar Nicholas II are 1, 3, 6, and 7.

(Record your answers in ascending order.)

Case Study #2: The Girl, the Army Sergeant, and the Bugs

Use the following list to answer the next question.

Observations of Insect Infestation upon the Remains of the Victim

1. Eggs
2. Beetles
3. Maggots
4. Adult Blowflies

①

2. Match an observation from the list above with each description given below

Insect Type:	<u>2</u>	<u>4</u>	<u>3</u>
Description:	not found on victim's body	laid eggs on victim's body	change into pupae

Case Study #3: The Farmhouse Murder

Use the following list to answer the next question.

Various Characteristics of Human Remains

1. Dismembered
2. Negroid Ancestry
3. Mongoloid Ancestry
4. Sexually assaulted
5. Died from stab wounds
6. Died from strangulation
7. Dead for less than 20 days
8. Dead for longer than 20 days

- ② 3. The characteristics numbered above that relate to the victim in this case study are 3, 4, 6, and 8.

(Record your answers in ascending order.)

Value
11

Part C: Written-Response Questions

Provide appropriate answers to the following. Explanations are best given in sentences.

Case Study #1: Bones that were Worth a Thousand Words

- ② 1. a. Identify two pieces of skull evidence that would have distinguished Czar Nicholas' wife, Czaritsa Aleksandra, from the adult male skeletons?
- Any two of the following are acceptable:**
- Supraorbital ridges in female forehead — slight
 - Few ridges below the eyes of a female
 - Female canine teeth — smaller
- ② b. Outline two pieces of evidence other than bone evidence used to identify the remains of Czaritsa Aleksandra.
- Mitochondrial DNA matched the British Royal family (1 mark)
 - Her unique dental work of platinum crowns (1 mark)

Case Study #2: The Girl, the Army Sergeant, and the Bugs

②

2. a. Explain why no beetles were found on the victim's skeletal remains.

- *The body had been dead for about 4 days. Beetles feed well after blowfly colonization that tends to take more than 20 days to occur. (1 mark)*
 - *Beetles feed upon mummified tissue that has less blood and body fluids, such as cartilage and bone. (1 mark)*
-
-

②

- b. Why did forensic entomologists in this case obtain data from a weather station near the crime scene?

- *The speed of insect colonization is greatly affected by weather. (1 mark)*
 - *The warmer the weather, the faster insect succession occurs. (1 mark)*
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Case Study #3: The Farmhouse Murder

③

3. List three pieces of evidence that suggest the victim had been dead for longer than 20 days.

Any three of the following are acceptable - 3 marks

- *Very little or no flesh on skeleton*
- *No blowflies found on the skeleton*
- *Beetles present on the remains*
- *Blowfly pupae membranes present. These take 16 to 26 days to appear.*

End of Assignment Booklet 6